

Course: Intelligent Agents and Multiagent Systems

Assignments 2025-2026

Students must deliver²:

- A report describing theory / algorithms, with evaluation/experimental results where it applies
- A presentation describing thoroughly theory / algorithms with specific examples, and with evaluation/experimental results, where it applies
- Code, where it applies
- **Students must declare their topic and team by a message sent via Lefkippos by December 20, 2025.**
- **1st version of the assignments will be discussed in a meeting in the week 27-31 January 2025 (the exact date will be announced in due time). This will provide feedback and steer students to be aligned with the objectives of the chosen topic.**
- **The final presentation will be given within the exams period i.e. by the Feb 15, 2025**

A. Distributed scheduling via auctions

3 persons: 50% theory and 50% implementation

- Specification of the distributed scheduling problem using agents' valuation functions
- Connection with competitive equilibria
- Specification and implementation of the ascending auction algorithm (convergence and termination)
- Experimental results

B. Congestion games

2 persons: 80% theory and 20% implementation

- Definitions and congestion games as potential games
- Non-atomic congestion games (NCG)
- Nash equilibria
- Specification and implementation of an algorithm for computing Nash equilibria in NCG
- The price of anarchy

C. Stochastic and dynamic games

2 persons: 70% theory and 30% implementation

- Definitions and examples of stochastic and dynamic games
- Computation of equilibria
- Specification and implementation of an algorithm for computing equilibria in a dynamic game

² Important notes:

4. Percentages show **rough** estimations of the expected distribution of effort to theory and implementation.
5. The number of persons per assignment is the maximum, however **I strongly discourage you working solo.**
6. Presentations and reports **should be of high quality and must reflect your deep understanding of the topic.** You do not need to add historical or other non-technical aspects in your reports & presentations.

D. Fictitious play and reinforcement learning for computing equilibria

2 persons: 30% theory and 70% implementation

- Repeated & zero sum (stochastic) games
- Fictitious play (FP) (theory and implementation)
- Reinforcement Learning (RL) (theory and implementation)
- Experimental results comparing FP and RL

E. DCOP

3 persons: 60% theory and 40% implementation

- Distributed Constraints Optimization Problems: Problem specification (discrete vs continuous)
- State of the art algorithms for discrete and continuous DCOP problems
- Experimenting with DPOP (implementation and results)

Starting point: <https://infoscience.epfl.ch/entities/publication/47a396f2-5563-4fec-b1fc-7f26ccbb9e4b/conferencedetails>

Basic material for all assignments is provided in

Y.Shoham and K.Layton-Brown, " Multiagent systems, Algorithmic, Game-Theoretic and Logical Foundations, Cambridge University Press, 2009.